

DANIEL J. VARON

Curriculum Vitae | July 2026

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EDUCATION

Ph.D., Atmospheric Chemistry , Harvard University	2015 – 2020
M.Sc., Applied Mathematics	
Secondary field in Computational Science & Engineering	
B.A., English Literature , McGill University	2010 – 2014
B.Sc., Physics , McGill University	2009 – 2014

PROFESSIONAL EXPERIENCE

Assistant Professor , Massachusetts Institute of Technology	2025 –
Boeing Career Development Professor of Aeronautics and Astronautics	
Department of Aeronautics and Astronautics	
Institute for Data, Systems, and Society	
Research Associate , Harvard University	2023 – 2025
School of Engineering and Applied Sciences	
Visiting Postdoctoral Research Associate , Princeton University	2021 – 2023
School of Public and International Affairs	
Postdoctoral Research Fellow , Harvard University	2020 – 2023
School of Engineering and Applied Sciences	

PUBLICATIONS (*SUBMITTED)

h-index = 31, total citations = 5062 (as of July 2026 on [Google Scholar](https://scholar.google.com/))

- *67. Perez-Carrasco, M., Nasr, M., Zhang, Z., Chulakadabba, A., Roger, J., Ottenheimer, R., Roche, S., Sargent, M., Chan Miller, C., **Varon, D. J.**, Warren, J., Guanter, L., Sun, K., Franklin, J., Chen, J., Garraffo, C., Liu, X., Gautam, R., and Wofsy, S.: Plume Segmentation from MethaneSAT with Cross-Sensor Transfer Learning and Physics-Informed Postprocessing, [preprint] submitted to *IEEE Trans. Geosci. Remote Sens.*, <https://doi.org/10.48550/arXiv.2605.24273>, 2026.
- *66. Gauld, K. D., **Varon, D. J.**, Thorpe, A. K., and Green, R. O.: Quantifying Nitrogen Oxide Point Sources with the EMIT Satellite Imaging Spectrometer, submitted to *Geophys. Res. Lett.*, 2026.
- *65. Hancock, S. E., Jacob, D. J., Jimenez, R., Ardila, A., Morales-Rincon, L., Rojas, N., Estrada, L. A., Balasus, N., East, J. D., Sulprizio, M. P., Wang, X., France, J. L., Potyk, L., Penn, E., Chen, Z., **Varon, D. J.**, Frankenberg, C., Baranski, M., Calcan, A., and Parker, R. J.: Applying Satellite Observations to Improve Bottom-Up National Emission Inventories for Methane: Application to Colombia, EGUsphere [preprint] *Atmos. Chem. Phys.*, <https://doi.org/10.5194/egusphere-2025-5478>, 2026.
- *64. Estrada, L. A., Jacob, D. J., He, M., East, J. D., **Varon, D. J.**, Balasus, N., Hancock, S. E., Sulprizio, M., Bowman, K. W., Worden, J. R., Reidy, E., and Runkle, B. R. K.: Quantifying national, state, and oil/gas field methane emissions and trends in the U.S. (2019-2024) through high resolution inversion of satellite observations, [preprint], <https://doi.org/10.5194/egusphere-2026-655>, 2026.
- *63. He, T.-L., **Varon, D. J.**, Kondragunta, S., Ren, X., Cohen, M. D., Carroll, B. J., Malarich, M., Peischl, J., de Jong, T. A., Gerritsen, J., Maasakkers, J. D., Cusworth, D. H., Duren, R. M., Brown,

- S. S., Warneke, C., Sweeney, C., Stratton, P., Brewer, A., Baidar, S., and Levin, E. J. T.: Coordinated satellite, aircraft, and ground-based observations of a large transient methane release, [preprint], <https://doi.org/10.31223/X54R07>, 2026.
- *62. Lama, S., Maasakkers, J. D., Zhang, X., Girard, M., Dutt, S., Nandgaonkar, S., **Varon, D. J.**, Sulprizio, M. P., Estrada, L. A., Balasus, N., Parker, R. J., Terao, Y., and Aben, I.: Evaluating national, state, and urban Indian methane emissions using satellites, EGU sphere [preprint], <https://doi.org/10.5194/egusphere-2025-6528>, 2026.
- *61. Chen, Z., Jacob, D. J., Lin, H., Balasus, N., Hancock, S. E., Estrada, L. A., East, J. D., Zhang, Y., Wang, X., He, M., Liu, M., and **Varon, D. J.**: Tropical wetland methane emissions and trends (2004–2023) inferred from Landsat-based inundated vegetation data, *Environ. Sci. Tech.*, in review, 2026.
- *60. Baray, S., Vogel, F., **Varon, D. J.**, Estrada, L. A., Balasus, N., Jones, D., McLinden, C., Griffin, D., Watine-Guiu, M., Duren, R., Cusworth, D. H., de Jong, T., and Maasakkers, J. D.: Monitoring methane emissions in Western Canada from space using a constellation of satellites, in review, 2026.
- *59. Bruno, J. H., Jacob, D. J., Wang, X., Sulprizio, M. P., Estrada, L. A., **Varon, D. J.**, Wofsy, S. C., Omara, M., and Gautam, R.: Integrating MethaneAIR aircraft and TROPOMI satellite observations in the Integrated Methane Inversion (IMI) to optimize methane emissions, [preprint], <https://doi.org/10.5194/egusphere-2025-4626>, 2026.
- *58. Sharma, S., Maasakkers, J. D., Dogniaux, M., Girard, M., McKeever, J. J., Jervis, D., Schuit, B., de Jong, T., Irakulis-Loitxate, I., Balasus, N., **Varon, D. J.**, and Aben, I.: Estimating methane emissions from surface coal mines using complementary satellite observations, *Environ. Sci. Tech.*, in review, 2026.
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35. **Varon, D. J.**, Jervis, D., Pandey, S., Gallardo, S. L., Balasus, N., Yang, L. H., and Jacob, D. J.: Quantifying NO_x point sources with Landsat and Sentinel-2 satellite observations of NO_2 plumes, *Proc. Natl. Acad. Sci.*, <https://www.pnas.org/doi/10.1073/pnas.2317077121>, 2024.
34. Nathan, B., Maasakkers, J. D., Naus, S., Gautam, R., Omara, M., **Varon, D. J.**, Sulprizio, M. P., Estrada, L. A., Lorente, A., Borsdorff, T., Parker, R. J., and Aben, I.: Assessing methane emissions from collapsing Venezuelan oil production using TROPOMI, *Atmos. Chem. Phys.*, 24, 6845–6863, <https://doi.org/10.5194/acp-24-6845-2024>, 2024.
33. Dogniaux, M., Maasakkers, J. D., **Varon, D. J.**, and Aben, I.: Report on Landsat 8 and Sentinel-2B observations of the Nord Stream 2 pipeline methane leak, *Atmos. Meas. Tech.*, 17, 2777–2787, <https://doi.org/10.5194/amt-17-2777-2024>, 2024.
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26. Pandey, S., van Nistelrooij, M., Maasakkers, J. D., Sutar, P., Houweling, S., **Varon, D. J.**, Tol, P., Gains, D., Worden, J., and Aben, I.: Daily detection and quantification of methane leaks using Sentinel-3: a tiered satellite observation approach with Sentinel-2 and Sentinel-5p, *Rem. Sens. Env.*, <https://doi.org/10.1016/j.rse.2023.113716>, 2023.
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dataset and deep learning benchmark for methane point source quantification using Sentinel-2 satellite imagery, *Rem. Sens. Env.*, <https://doi.org/10.1016/j.rse.2023.113708>, 2023. [PDF]

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Selected as Highlight Paper

23. Chen, Z., Jacob, D. J., Gautam, R., Omara, M., Stavins, R. N., Stowe, R. C., Nesser, H., Sulprizio, M. P., Lorente, A., **Varon, D. J.**, Lu, X., Shen, L., Qu, Z., Pendergrass, D. C., and Hancock, S.: Satellite quantification of methane emissions and oil-gas methane intensities from individual countries in the Middle East and North Africa: implications for climate action, *Atmos. Chem. Phys.*, 23, 5945–5967, <https://doi.org/10.5194/acp-23-5945-2023>, 2023.
22. Lu, X., Jacob, D. J., Zhang, Y., Shen, L., Sulprizio, M. P., Maasakkers, J. D., **Varon, D. J.**, Qu, Z., Chen, Z., Hmiel, B., Parker, R. J., Boesch, H., Wang, H., He, C., and Fan, S.: Observation-derived 2010–2019 trends in methane emissions and intensities from US oil and gas fields tied to activity metrics, *Proc. Natl. Acad. Sci.*, <https://doi.org/10.1073/pnas.2217900120> 2023.
21. Gorroño, J., **Varon, D. J.**, Irakulis-Loitxate, I., and Guanter, L.: Understanding the potential of Sentinel-2 for monitoring methane point emissions, *Atmos. Meas. Tech.*, 16, 89–107, <https://doi.org/10.5194/amt-16-89-2023>, 2023.
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17. Qu, Z., Jacob, D. J., Zhang, Y., Shen, L., **Varon, D. J.**, Lu, X., Scarpelli, T., Bloom, A., Worden, J., and Parker, R. J.: Attribution of the 2020 surge in atmospheric methane by inverse analysis of GOSAT observations, *Environ. Res. Lett.*, 17, 9, <https://doi.org/10.1088/1748-9326/ac8754>, 2022.
16. Maasakkers, J. D., **Varon, D. J.**, Elfarsdóttir, A., McKeever, J., Jervis, D., Mahapatra, G., Pandey, S., Lorente, A., Borsdorff, T., Foorthuis, L. R., Schuit, B. J., Tol, P., van Kempen, T. A., van Hees, R., and Aben, I.: Using satellites to uncover large methane emissions from landfills, *Sci. Adv.*, 8, 32, <https://doi.org/10.1126/sciadv.abn9683>, 2022.
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 5. Zhang, Y., Gautam, R., Pandey, S., Omara, M., Maasakkers, J. D., Sadavarte, P., Lyon, D., Nesser, H., Sulprizio, M. P., **Varon, D. J.**, Zhang, R., Houweling, S., Zavala-Araiza, D., Alvarez, R. A., Lorente, A., Hamburg, S. P., Aben, I., and Jacob, D. J.: Quantifying methane emissions from the largest oil producing basin in the U.S. from space. *Science Advances*, 6, 17, <https://www.science.org/doi/10.1126/sciadv.aaz51>, 2020.
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2. **Varon, D. J.**, Jacob, D. J., McKeever, J., Jervis, D., Durak, B. O. A., Xia, Y., and Huang, Y.: Quantifying methane point sources from fine-scale satellite observations of atmospheric methane plumes. *Atmos. Meas. Tech.*, 11, 5673–5686, <https://doi.org/10.5194/amt-11-5673-2018>, 2018.

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MENTORING

Postdocs

- Tailong He (MIT, Harvard), 2024–

Graduate students

- Lucas Estrada (PhD, Harvard), 2025–
- Krishma Mehta Rajesh (PhD, MIT), 2025–
- Vassiliki Mancoridis (PhD, MIT), 2025–
- Kevin Gauld (PhD, MIT), 2025–
- François Martin-Monier (MSc, ETH Zürich), 2023.
- Marc Watine (MSc, ETH Zürich), 2023.

Undergraduate students

- Jayden Graham (MIT), 2026.
- Asael (Acosta), 2025.
- Khue Phan (MIT), 2025.
- Israel Garcia (MIT), 2025.
- Chevaugn Campbell (Kenyon College), 2022.
- Daniel Shen (Harvard University), 2021.

SERVICE

Professional Service

- | | |
|-------|--|
| 2025– | Model Scientist, Integrated Methane Inversion (IMI) |
| 2025– | Co-Nested-Model Scientist, GEOS-Chem |
| 2024– | Scientific Advisory Board, Methane Emissions Detection Using Satellites Assessment (MEDUSA) project (SRON Space Research Organisation Netherlands) |
| 2023– | Associate Editor, <i>Atmospheric Measurement Techniques</i> |